MODELLING THE IMPACT OF LOWERED ATMOSPHERIC NITROGEN DEPOSITION ON A NITROGEN SATURATED FOREST ECOSYSTEM

C. J. KOOPMANS[1,*] and D. VAN DAM[2]
[1] *Landscape and Environmental Research Group, University of Amsterdam, Nieuwe Prinsengracht 130, 1018 VZ Amsterdam, The Netherlands;* [2] *Department of Soil Science and Geology, Wageningen Agricultural University, P.O. Box 37, 6700 AA Wageningen, The Netherlands*
E-mail: mfai@igc.apc.org

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Abstract. For a Douglas fir forest ecosystem subjected to an experimental decrease in nitrogen (N) deposition, N dynamics were simulated using the dynamic simulation model NICCCE. Meteorological driving variables and N concentrations in throughfall were input to the model, that simulated results of a ^{15}N tracer experiment, C and N concentrations in the soil, soil water chemistry and tree biomass. Four years of ambient N deposition, followed by four years of N deposition manipulations by means of a roof construction beneath the forest canopy, were modelled. Simulation of this second period was performed for a high-N treatment (37 kg N ha^{-1} yr^{-1}) and a low-N treatment with throughfall-N at natural background level (6 kg N ha^{-1} yr^{-1}). Calibration and model performance is discussed and compared to results of field experiments. The quick response of soil water chemistry after lowering N deposition and the ^{15}N tracer signal observed in soil water at 90 cm soil depth, were simulated closely by the calibrated model. $^{15}NH_4$-N data could only be simulated by accounting for bypass flow, indicating that throughfall water did not fully interact with the soil. Using the calibrated parameter set of the low-N treatment for the high-N treatment resulted in a lower model performance, although time trends were reproduced well also for this treatment. A sensitivity analysis showed model outcome of N transformations to be very sensitive to soil microbial parameters, such as the C efficiency. Use of the ^{15}N tracer data in the calibration lowered uncertainties of these sensitive model parameters. Evaluation of the N input-output budget and microbial N transformations in the ecosystem revealed that lowering N inputs in this N saturated forest soil resulted in a more than proportional decrease of N leaching losses out of the soil system. Gross N transformations decreased under lowered N input, in particular the formation of NO_3-N. Net N mineralization was not affected after four years of N manipulations. Net nitrification was decreased to about one third of the rate observed at the high-N deposition plot. Combining ^{15}N tracer data with dynamic simulation modelling provides a powerful tool to improve model performance and process descriptions, and to evaluate impacts of atmospheric N deposition on N cycling in ecosystems.

Key words: coniferous ecosystems, Douglas fir, ^{15}N, NICCCE, NITREX, nitrogen, simulation model

1. Introduction

Elevated atmospheric nitrogen (N) deposition levels, in many parts of Europe and North America, are recognized to cause N saturation of forested ecosystems (Aber *et al.*, 1989; Gundersen, 1992). Nitrogen saturation as used here refers to the condition in which N availability exceeds the capacity of plants and soil microbes

[*] Present address: Michael Fields Agricultural Institute, W2493 County Road ES, WI 53120, USA

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to accumulate N (Aber *et al*., 1989; Gundersen, 1992). Accompanying effects like increased nitrate (NO_3) leaching (Kahl *et al*., 1993), soil acidification (Van Breemen and Verstraten, 1991) and forest dieback (Schulze, 1989) are seen as major environmental hazards. Within the European Union project NITREX (NITRogen saturation EXperiments), N saturation of forested ecosystems and its reversibility are evaluated in eight European forests along a gradient of N deposition by means of field-scale manipulation experiments (Wright and Van Breemen, 1995). In these experiments, questions on the impact of N deposition levels on N cycling and the effectiveness of abatement strategies are studied. The goal is to increase our understanding of biogeochemical processes related to N transformations and the dependence on atmospheric N inputs.

Most of the forest ecosystems in the Netherlands can be characterized as N saturated due to forty years of elevated N deposition (Van Breemen and Verstraten, 1991). To investigate whether a decrease in atmospheric N inputs can reverse N saturation and its accompanying effects, a field-scale manipulation experiment was carried out in a N-saturated Douglas fir forest in the Netherlands (Boxman *et al*., 1995). By means of a roof construction beneath the canopy, two experimental plots with respectively high (37 kg N ha^{-1} yr^{-1}) and strongly attenuated (6 kg N ha^{-1} yr^{-1}) N deposition were established. The fate into the most important N pools of the deposited NH_4-N and the response of individual ecosystem pools and processes to drastically lowered N deposition were examined by adding ^{15}N as $(^{15}NH_4)_2SO_4$ for one year to the throughfall water at the two plots (Koopmans *et al*., 1996). The model NICCCE (Nitrogen Isotopes and Carbon Cycling in Coniferous Ecosystems) was developed within the NITREX project (Van Dam and Van Breemen, 1995) as a tool to interpret the experimental results of N, carbon (C) and isotope cycling in coniferous forest ecosystems.

In the present study we discuss the calibration and application of the NICCCE model using the biogeochemical data available from the N-saturated Douglas fir forest and data from the ^{15}N tracer experiment. The response of the Douglas fir ecosystem to a sudden experimental decrease of the atmospheric N input is described and the model output is compared to field data, to evaluate process descriptions, parameter values and N fluxes.

2. Materials and Methods

2.1. The NICCCE model

The process-oriented model NICCCE includes heat and water transport, primary production, decomposition, mineralization, secondary production and isotope cycling of C and N in a mature coniferous forest stand, on a one-dimensional multicompartment soil profile (Van Dam and Van Breemen, 1995). Only a brief description of those components of the model relevant to this study is presented here since the model has been described in full detail earlier (Van Dam and Van Breemen, 1995).

In the *primary production* module, the forest canopy is divided into needles (5 age-classes), branches, stems (sapwood and heartwood), fine (<2 mm) and coarse roots. Five crown positions, differing in leaf area index are distinguished in the canopy. Photosynthesis is integrated over these crown positions for five age-classes of needles. Maximum photosynthetic capacity depends on the N content of the foliage. Carbon and N allocation to biomass compartments are calculated using allocation parameters and a C demand function depending on the N content of foliage and fine roots.

In the *decomposition* module, tree litter is divided into four components (polysaccharides, proteins, hemicellulose and lignin) and decomposed by microbes, according to first order kinetics. Microbial litter is further decomposed into a 'structural' part (turnover 1–2 yr) mainly consisting of cell wall materials. This fraction turns into 'humified' (turnover 10–20 yr), 'stable' (turnover 50–100 yr) and 'resistant' (turnover 500–2000 yr) organic matter.

The microbial transformations of N and C are explicitly simulated within NICCCE with a principle of substrate-use efficiencies. The substrate-use efficiency of C (N) is defined as the amount of C (N) assimilated by microbial biomass per unit of C (N) in the substrate that is consumed (Van Dam and Van Breemen, 1995). The substrate-use efficiencies depend on C:N ratios of the microbial biomass. A sensitivity analysis of the NICCCE model revealed a strong influence of the microbial C-use efficiency on N input-output budgets (Van Dam and Van Breemen, 1995). In the *mineralization* process, the substrate-use efficiency of organic N was defined as the fraction of organic N in the decomposing substrate pool that is incorporated into microbial biomass, whereas the remaining part is dissimilated as NH_4. The same principle was used in the *nitrification* process for NH_4 assimilation and dissimilation of NO_3, and for C assimilation and dissimilation of CO_2. Due to this principle, nitrification is low in forest ecosystems with low NH_4 availability and high C:N ratio of the substrate, whereas in N saturated forest ecosystems low C:N ratios result in a considerable nitrification rate. Autotrophic and heterotrophic nitrification are lumped in this principle and therefore described with the same physiological response. *Denitrification* is also included in the model as a potential denitrification rate corrected for soil moisture, temperature and NO_3 concentration based on Michaelis-Menten kinetics.

NICCCE simulates isotope cycling of ^{14}N, ^{15}N, ^{12}C, ^{13}C and ^{14}C. Although isotope fractionation is included in the model (Van Dam and Van Breemen, 1995) it is insignificant in the present model evaluation, as in the tracer experiment all enrichments were well above the natural abundance levels (Koopmans *et al.*, 1996). ^{15}N fluxes (J_{15N}) are calculated as $J_{15N} = K*[^{14}N+^{15}N]*(1-\alpha)*R$ and ^{14}N fluxes as $J_{14N} = K*[^{14}N + ^{15}N]*(1-R)$, in which K is the reaction rate constant, α is an isotope discrimination factor and R is the isotope ratio ($^{14}N/(^{14}N + ^{15}N)$).

Additional information on process formulations of forest hydrology, heat transport, compartmentalization, boundary conditions and time steps used in NICCCE, can be found in Van Dam and Van Breemen (1995). Input data for the model includes atmospheric N input levels as throughfall-N concentration and canopy exchange. The precipitation flux, temperature, short wave radiation, relative humidity, wind speed and atmospheric CO_2 levels are used as driving variables.

2.2. RESEARCH AREA

The Douglas fir (*Pseudotsuga menziesii* (Mirb.) Franco.) forest is located in the central part of the Netherlands, about 125 km east of the North Sea at latitude 52°13′ N, and longitude 5°39′ E near the village Speuld. The forest is a 2.5 ha site in a 50 km^2 forested area. The forest is reasonably homogenous with a density of 800 trees ha^{-1}. The firs are 35-years old, about 22 m high with a leaf area index of 8–11 m^2 m^{-2}. The trees have a high foliage mass (10–11 tons ha^{-1}) with a high N concentration (1.6–2.5%). Total stemwood production is over 5 tons ha^{-1} yr^{-1}.

The trees were planted on sandy loam and loamy sand, well drained, with a ground water table at about 40 m. A 3 to 7 cm thick organic layer has developed on the mineral soil, which was classified as a Haplic Podzol (FAO/UNESCO, 1988). Soil organic C in the mineral soil ranges from 4.7% at 0–10 cm depth decreasing to 0.3% at 90 depth. The pH_{CaCl2} increases with soil depth from 3.2 to 5.4. Effective CEC ranges from 28 to 105 mmol$_c$ kg^{-1} and base saturation is below 1%. C:N ratios in the organic layers are in the range of 21–23, increasing in the mineral soil up to 30. The mean annual air temperature is 9.3 °C, the mean annual precipitation amounts to 800 mm. Atmospheric N deposition in bulk precipitation (22 kg N ha^{-1} yr^{-1}) increases in throughfall to approximately 50 kg N ha^{-1} yr^{-1}, mainly as NH_4 (74%) (Boxman *et al*., 1995).

2.3. MANIPULATIONS

In May 1989 a field scale manipulation experiment was started at the Speuld site. In order to manipulate the N input, a transparent roof was build under the canopy at 2 to 3 m height, covering two manipulated plots. A control plot was located outside the roof. The 100 m^2 plots manipulated (with bufferstrips of 2 m around each plot) received either a high inorganic N load (37 kg N ha^{-1} yr^{-1}) or a strongly attenuated inorganic N load (6 kg N ha^{-1} yr^{-1}) by the addition of an artificial throughfall solution to the low-N deposition plot. Nitrogen losses during collection and storage of ambient throughfall resulted in the apparent discrepancy between N addition to the high-N plot (37 kg N ha^{-1} yr^{-1}) and under ambient conditions reported by Boxman *et al*. (1995). Monitoring of soil water chemistry, foliage composition, water potentials, soil and air temperatures was continued at all plots until 1995. Details can be found in Boxman *et al*. (1995) and Koopmans *et al*. (1996).

With an objective of following the fate of the deposited NH_4-N at both deposition levels, a field-scale ^{15}N tracer experiment was carried out in this forest (Koopmans

Table I

Initial N and C pools of the Douglas fir trees at Speuld (adapted from Van Dam and Van Breemen, 1995; see text for data sources)

	Needles	Branches	Stems	Roots fine[a]	Roots coarse[b]
N pool (kg ha^{-1})	180	110	175	12	40
C pool (kg ha^{-1})	5100	16150	11300	1600	8000

[a] Diameter <2 mm.
[b] Diameter >2 mm.

et al., 1996). For one year (May 1992 to May 1993), the stable isotope ^{15}N was applied as 99.4% enriched $(^{15}NH_4)_2SO_4$ on the manipulated plots. During that year, a total amount of 0.69 and 0.64 mg m^{-2} ^{15}N was added with the throughfall water to the high-N and low-N deposition plots, respectively. Major ecosystem pools and fluxes were sampled on a regular basis for determination of total N and ^{15}N concentrations (Koopmans *et al.*, 1996). Vegetation and soils were sampled in February 1992, 1993 and 1994. From February 1992 to February 1994 the soil water was sampled every fortnight from eight replicate ceramic lysimeter cups per plot installed at 0, 10, 50 and 90 cm depth. Samples were pooled for each depth on volume-basis for three-month intervals. NH_4 and NO_3 concentrations were analyzed on an autoanalyzer. NH_4 and NO_3 were separated using the micro-diffusion technique prior to ^{15}N analysis on a mass-spectrometer (Koopmans *et al.*, 1996).

Because of relatively low ^{15}N enrichments in the manipulations, results are expressed in $\delta^{15}N$ notation. With R_{sample} and $R_{standard}$ representing atom% ^{15}N of the sample and standard (atmospheric N_2 with 0.3663 atom% ^{15}N), respectively, the per mil ^{15}N excess (‰) is defined by:

$$\delta^{15}N = (R_{sample}/R_{standard} - 1) \times 1000‰$$

2.4. Input Data

NICCCE was applied to the Speuld site for the period 1987 to 1995. For the years 1987 to 1989 an extensive data set on atmospheric deposition, soil and soil water chemistry was available from Van der Maas (1990). Initial soil conditions were based on this data. Soil physical characteristics (water retention characteristics and unsaturated hydraulic conditivity) were obtained from Tiktak *et al.* (1988). Initial biomass and root distributions were available from investigations of Evers *et al.* (1991), Olsthoorn (1991) and Olsthoorn and Tiktak (1991) (Tables I and II).

For the 1990 to 1995 period, results on the manipulated plots presented by Boxman *et al.* (1995) and Koopmans *et al.* (1996) were used. All plots were located in the same research area (1 ha) but were subject to some spatial soil variability. Observed $^{15}NO_3$ and $^{15}NH_4$ concentrations in soil water refer to the

Table II

Fixed root uptake distribution and initial soil organic matter pools for the Douglas fir ecosystem in the NICCCE simulations (see text for references). Organic matter is divided in 'structural' (turnover 1.5 yr), 'humified' (turnover 11.5 yr), 'stable' (turnover 130 yr) and 'resistant' (turnover 1200 yr) pools

Depth	Roots	Structural		Humified		Stable		Resistant	
		C:N	N	C:N	N	C:N	N	C:N	N
(cm)	(%)	(–)	(kg ha^{-1})	(–)	(kg ha^{-1})	(–)	(kg ha^{-1})	(–)	(kg ha^{-1})
+5–0	15.61	18.0	300.0	18.0	206.0	18.0	62.2	19.0	8.3
0–9	22.05	20.0	130.0	21.0	119.0	21.0	228.6	22.0	344.5
9–18	15.12	20.8	72.0	22.1	47.5	22.0	101.8	23.0	169.1
18–30	12.88	21.9	64.0	23.0	46.7	23.0	109.6	24.0	198.3
30–50	13.90	22.1	68.0	23.0	56.0	23.0	142.2	24.0	279.6
50–70	10.34	22.2	54.0	25.0	44.8	25.0	119.2	25.0	256.4
70–100	6.32	22.8	46.0	26.0	55.3	27.0	145.9	26.1	333.2

Table III

In the calibration of the NICCCE model, model parameters were optimized on the basis of least sum of squared residuals (SSR) between simulated and measured values of some physical, biogeochemical and ^{15}N isotope characteristics

Physical state variables
Soil temperature at 90 and 200 cm
Throughfall fluxes
Soil water potentials (pF; 90 cm)
Biogeochemical state variables
NO_3 concentrations in soil water at 5 depths (0, 10, 25, 50 and 90 cm)
NH_4 concentrations in soil water at 5 depths (0, 10, 25, 50 and 90 cm;
Cl concentrations in soil water at 5 depths (0, 10, 25, 50 and 90 cm)
N percentages in 1st-year needles
Total-C in needles
Total-C in the stem
Isotope ratios
$\delta^{15}NO_3$-N values in soil water (90 cm)
$\delta^{15}NH_4$-N values in soil water (90 cm)
δ^{15}N values in the 1st-year needles
δ^{15}N values in the organic layer
δ^{15}N values in the mineral soil (0–10 cm)

samples pooled for three-month periods. Data on total N, ^{15}N and C in the needles and stem were linearly interpolated between sampling dates.

2.5. Model calibration and performance

Model parameters were optimized on the basis of least sum of squared residuals (SSR) between simulated and measured values of important physical and biogeochemical characteristics (Table III). All individual SSR's were scaled so that each SSR contributed approximately equally to the sum. All 109 parameters of the model were varied at random for a large number of runs (several hundreds). In a second step of the optimization, less sensitive parameters (51) were fixed. The remaining set of parameters (58) was optimized using the Simplex algorithm (Nelder and Mead, 1965; Freijer, 1990), resulting in a convergence towards a minimum SSR. The Simplex optimization algorithm is especially suitable for optimizing parameters in complex simulation models, because it is not sensitive to local minima and also handles non-linear functions.

Model performance was visually (qualitative) evaluated as well as by a number of quantitative (statistical) measures described in detail by Janssen and Heuberger (1995). The bias between average values of model predictions and observations is expressed by the Average Error (AE). A positive AE indicates that model simulations overestimate observations, whereas a negative AE indicates that the model underestimates the observed values. The Normalized Root Mean Square Error (RMSE) resembles a coefficient of variation of the discrepancies between simulated and observed values around the mean of the observations. The Modelling Efficiency (ME) quantifies the improvement of the simulation model over a simple model represented by the mean of all observations (Janssen and Heuberger, 1995). A ME close to 1 indicates a perfect fit of the simulations to the observations whereas a ME efficiency close to 0 or smaller indicates no improvement of the simulation model in comparison with a simple model represented by the mean of all observations. Of course, such a simple model does not include any mechanistic interpretation.

We also applied the model to the high-N plot, using the same optimized parameter set as generated for the low-N plot, to evaluate model performance under conditions of continuous high N deposition levels. An ideal parameter set should describe both treatments. However, it should be noticed that this application to the high-N plot is not an independent validation of the model, as it includes the same pre-treatment period (1987 to 1989) used for the low-N plot.

2.6. Bypass flow

Bypass flow was included as an option in the model in order to achieve a better fit for some observations. A fraction of the throughfall infiltrated directly to the soil water compartments at different depths and was then mixed with these compartments. The fraction was obtained by calibration. Simulations including bypass flow were compared to simulations without any bypass flow. Results are discussed in more detail with results on $^{15}NH_4$-N.

2.7. Sensitivity analysis

To investigate the impact of the ^{15}N tracer data on overall model performance and in particular on model parameter uncertainties, a sensitivity analysis was performed. Based on this sensitivity analysis and using Kolmogorov statistics (Spear and Hornberger, 1980; Davis, 1986) we were able to identify those model parameters whose uncertainty was reduced by the use of ^{15}N tracer data in the simulations.

After calibration of the model for the low-N plot, model parameters (58) were varied at random (by 8%). We distinguished two *cases* to accept or reject runs (Spear and Hornberger, 1980). In *case 1* no ^{15}N tracer data are available. In *case 2* ^{15}N tracer data are available. In *case 1*, runs were accepted if the SSR's between simulated and measured values of the NO_3 concentrations in soil water and N% in needles remained within twice the standard deviation of the mean for all runs. In *case 2* runs were accepted if the two criteria of case 1 were fulfilled, and if in addition SSR's between simulated and measured values of $\delta^{15}NO_3$-N in soil water and $\delta^{15}N$ in the needles remained within twice their standard deviation. For both cases sensitivities of all model parameters were ranked according their Kolmogorov d-statistic, i.e. the maximum difference between cumulative probability distributions of accepted and rejected runs (Spear and Hornberger, 1980; Davis, 1986). The difference in accepted runs between *case 1* and *2* may be considered to result from some 'wrongly accepted' runs in situations without information on tracers (*case 1*). According to the same concept the maximum difference between cumulative probability distributions of accepted (*case 1*) and 'wrongly accepted' runs indicates which model parameters are most sensitive to information provided by the ^{15}N tracer data (*tracer effect*). Model parameter ranking according to this principle therefore indicates where incorporation of ^{15}N tracer data reduces the uncertainty in model parameters.

3. Results

3.1. Soil temperature, soil water potential and chloride

Average differences between observed and simulated soil temperature (90 and 200 cm depth) were less than 0.2 °C (not shown). Figures 1 and 2 and Table IV show model performance for soil water potentials and Cl concentrations. The response of soil water potentials at 90 cm soil depth was somewhat delayed in the simulations, and high soil water potentials in summer were underestimated by the model. The lower soil water potentials during winter were simulated well.

Chloride was assumed to act as a conservative tracer. Although the NICCCE simulations followed the experimental Cl concentrations (Figure 2), some extremely high Cl concentrations observed (particularly at 90 cm depth) were not reached in the simulations. These high Cl concentrations contradicted observations at 0 cm depth (below the organic layer), as they could not be explained by decreasing soil

Table IV
Performance statistics of the calibrated model for the low-N and high-N plot at Speuld

	Unit	Obs.	Low-N plot				High-N plot			
		no.[a]	MO[b]	AE[c]	NRMSE[d]	ME[e]	MO[b]	AE[c]	NRMSE[d]	ME[e]
Soil water potentials[f]	(log(-cm))	1285	1.97	–0.09	0.24	0.36	1.97	–0.08	0.24	0.37
Cl	(μmol L^{-1})	636	624.7	–120.2	0.57	–0.01	550.1	–31.3	0.56	–0.20
NO_3	(μmol L^{-1})	655	538.1	–51.0	0.80	0.54	859.3	22.1	0.59	–0.12
NH_4	(μmol L^{-1})	655	51.6	–7.9	1.68	0.99	82.8	–30.6	1.68	0.34
N needles	(%)	167	1.79	–0.02	0.03	0.56	3.65	–0.03	0.02	
C needles	(Mg ha^{-1})	167	47.9	0.07	0.05	0.69	47.9	1.57	0.07	0.53
C stem	(Mg ha^{-1})	167	126.5	–0.08	0.04	1.0	126.5	–0.28	0.01	0.99
$\delta^{15}NO_3$	(‰)	46	380.9	–79.8	0.55	0.71	340.5	–2.42	0.54	–0.07
$\delta^{15}NH_4$	(‰)	46	362.4	–116.6	1.16	–0.05	362.2	–82.9	0.84	–0.92
δ^{15}N needles	(‰)	33	34.0	–2.08	0.22	0.88	28.7	16.3	0.75	–0.43
δ^{15}N organic	(‰)	8	10.0	2.14	1.59	0.74	2.39	12.5	17.8	–0.44

[a] Number of data used as observations in NICCCE. [b] Mean of observations. [c] Average error. [d] Normalized root mean square error.
[e] Modelling efficiency (Janssen and Heuberger, 1995). [f] For both plots the same soil water potentials were used in the simulations.

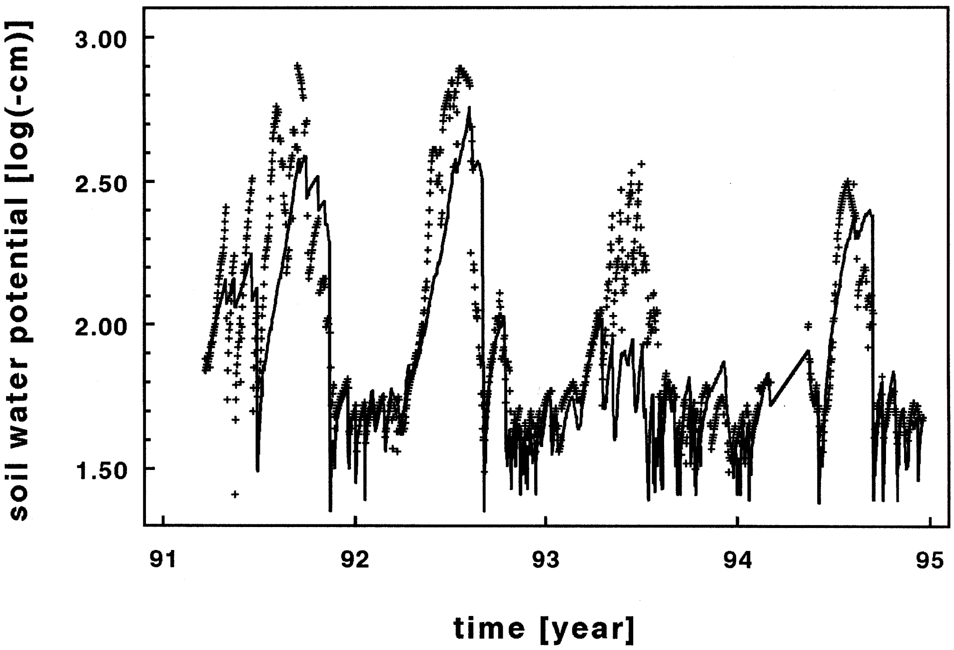

Figure 1. Observed (symbols) and simulated (line) soil water potentials at 90 cm soil depth at the plots under the roof.

water fluxes due to evapotranspiration. The model performance is reflected in the negative AE for Cl (Table IV). The ME close to zero indicates that the simulations describe these Cl concentrations hardly better than a model represented by the mean of all observations. This is caused by the wide range of scatter in the data.

3.2. Nitrate and Ammonium

Observed and simulated NO_3 concentrations for the manipulated plots are presented in Figure 3. Simulated NO_3 concentrations followed closely the observed concentrations throughout the profile. After lowering atmospheric N deposition, the observed NO_3 concentrations decreased rapidly in the low-N plot. The calibration revealed that this quick response in the NO_3 concentrations could be satisfactorily simulated. Using the same parameter-set for the high-N plot, with an identical pre-treatment period (1987 to 1990), showed that NO_3 concentrations were somewhat overestimated in the upper soil at this plot (AE>0, Table IV), whereas simulations followed the observed NO_3 concentration at 90 cm soil depth. The ME of 0.54 for the low-N plot indicates a good model performance, whereas a much lower ME (0.12) was found for the high-N plot.

The observed ammonium concentration decreased rapidly with depth at both plots (Figure 4). In the pre-treatment period, NH_4 concentrations were underestimated in the simulations for the upper soil and somewhat overestimated in the

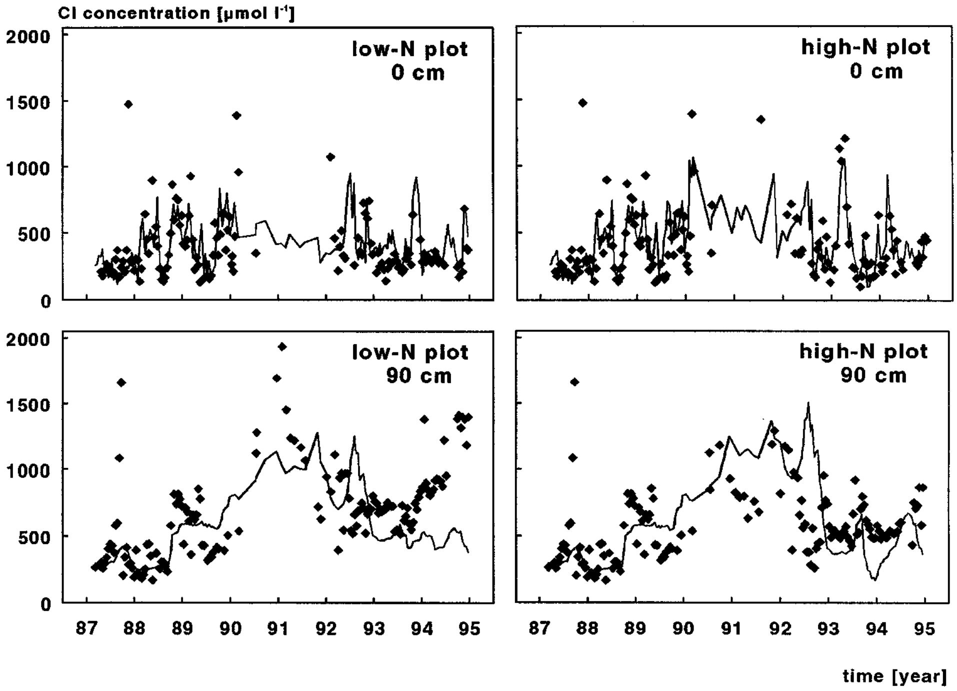

Figure 2. Observed (symbols) and simulated (line) Cl concentrations in soil water below the organic layer (0 cm) and at 90 cm soil depth during ambient atmospheric N deposition 1987–1989 (both series) and in the low-N and high-N plot at Speuld during 1990–1995.

deeper soil. After the roof was erected, observed NH_4 concentrations in the upper soil decreased at the low-N plot. This decrease was inadequately simulated by the model. At the high-N plot, NH_4 concentrations were reasonably simulated for the upper soil. At 50 cm, some high NH_4 concentrations were not fully described by the model.

3.3. ISOTOPE RATIOS

The simulated $^{15}NO_3$ levels at 90 cm followed the observed isotope levels in the low-N plot reasonably well (Figure 5 and Table IV).

The peak value observed in the second three-month period was not reached completely in the simulation. Simulated isotope levels decreased somewhat quicker than the isotope levels observed in the field. Compared to observations, simulated $^{15}NO_3$ concentrations for the high-N plot were delayed and ultimately showed a lower response. In this high-N plot, the highest $\delta^{15}N$ values were overestimated in the simulation when compared to field observations. The ME was considerably better for the low-N as compared to the high-N plot (Table IV). The $^{15}NH_4$ levels in the soil water in the ceramic cups at 90 cm soil depth suggested that ^{15}N reached these cups within the first three months after the start of the ^{15}N experi-

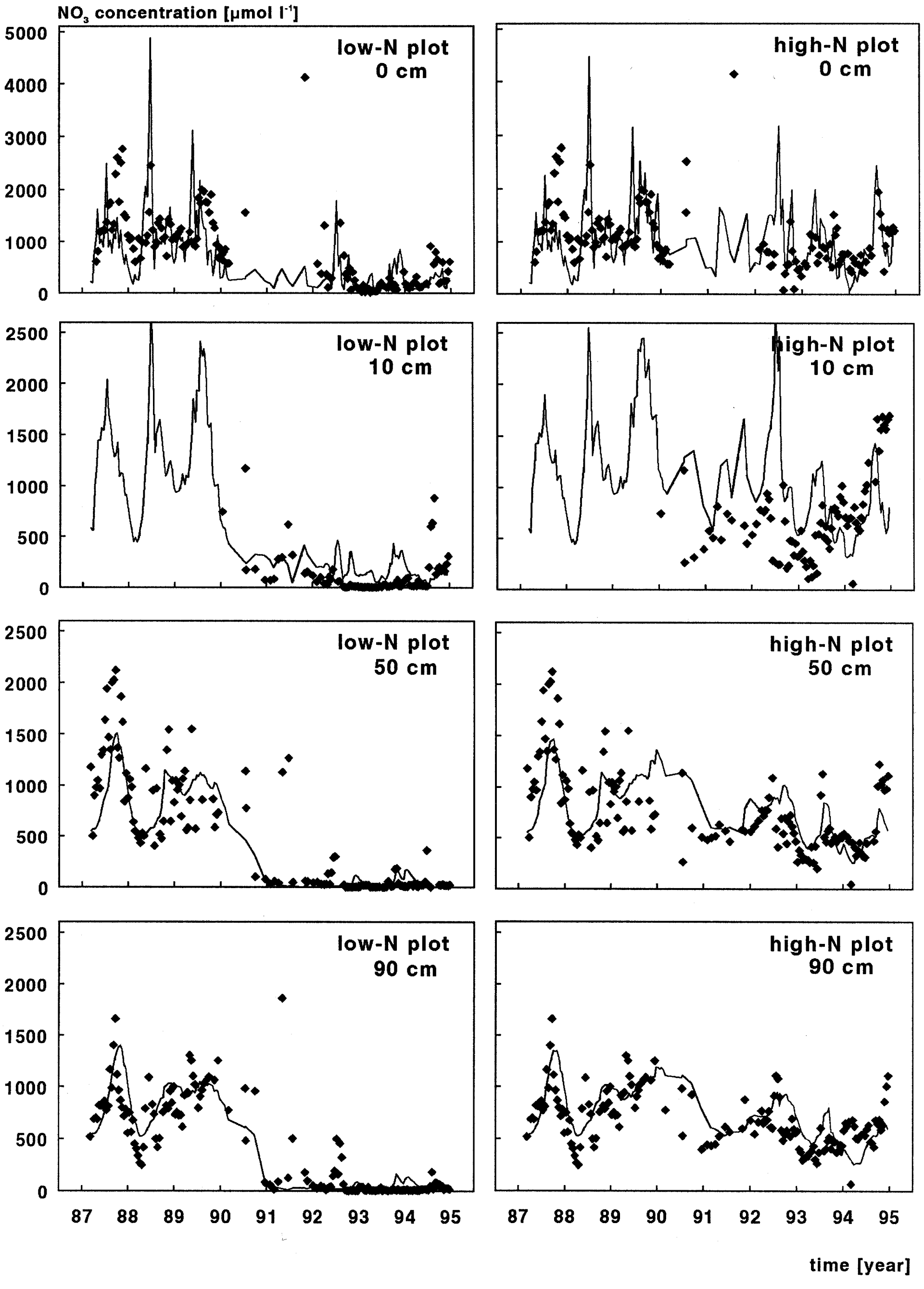

Figure 3. Observed (symbols) and simulated (line) NO_3 concentrations in soil water below the organic layer (0 cm) and at 10, 50 and 90 cm soil depth during ambient atmospheric N deposition 1987–1989 (both series) and in the low-N and high-N plot at Speuld during 1990–1995.

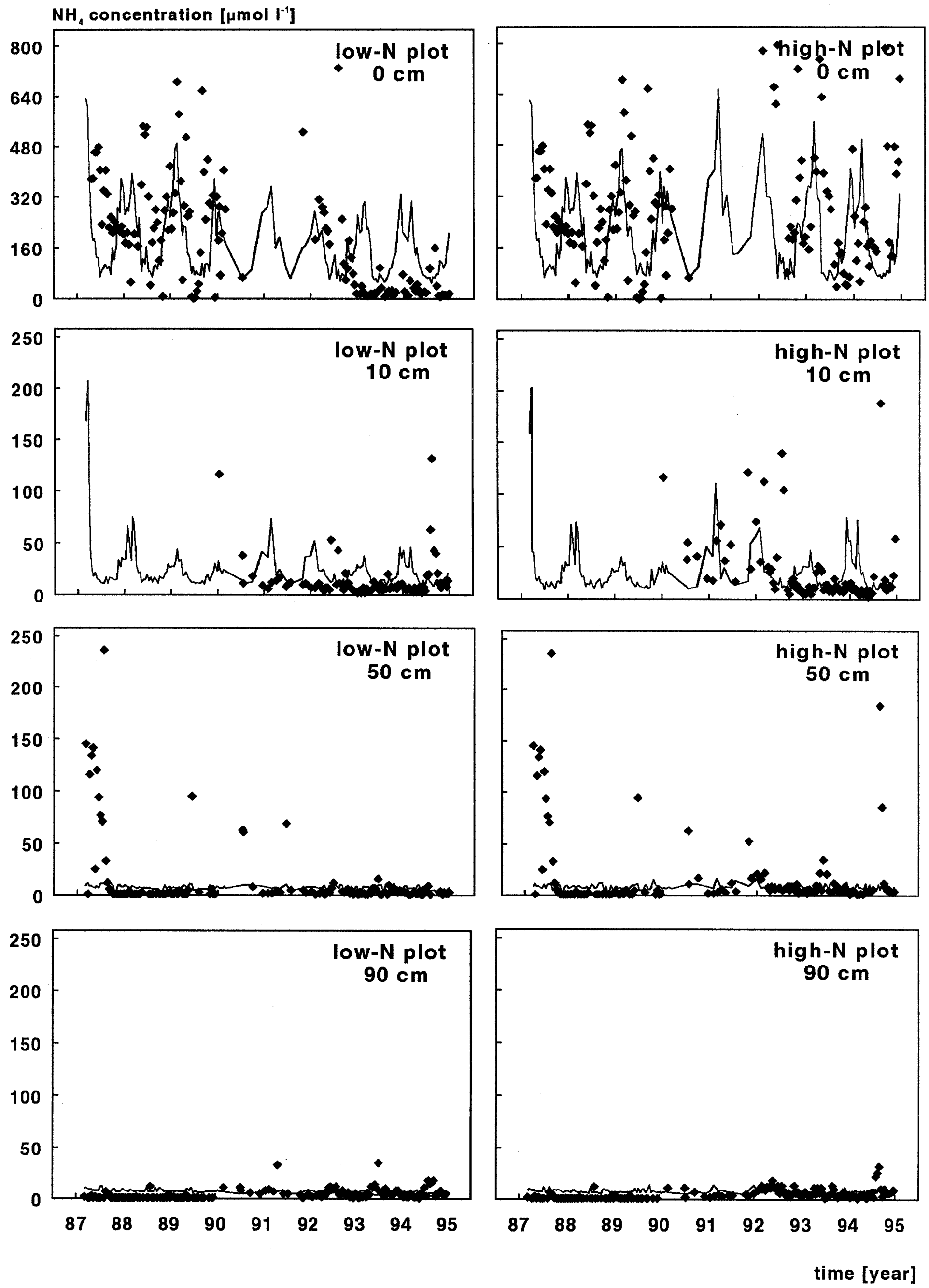

Figure 4. Observed (symbols) and simulated (line) NH_4 concentrations in soil water below the organic layer (0 cm) and at 10, 50 and 90 cm soil depth during ambient atmospheric N deposition 1987–1989 (both series) and in the low-N and high-N plot at Speuld during 1990–1995.

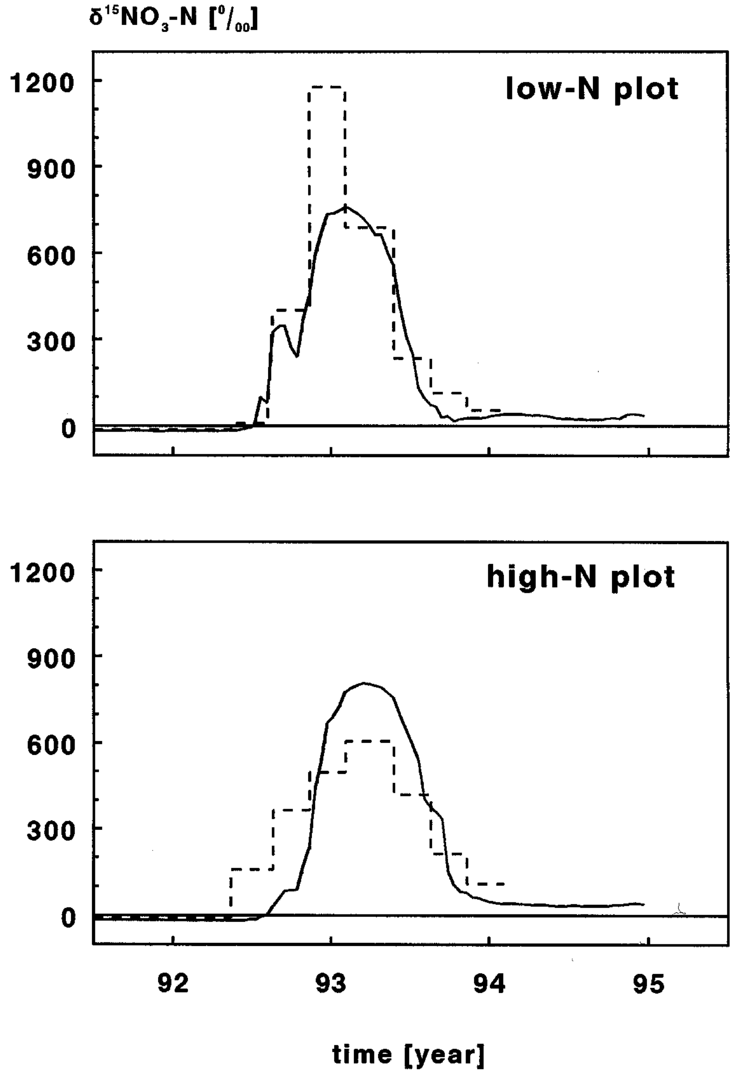

Figure 5. Observed (dashed line) and simulated (solid line) $\delta^{15}NO_3$ values in soil water at 90 cm soil depth in the low-N and high-N plot at Speuld.

ment (Figure 6). With the NICCCE model it seemed impossible to simulate such a rapid enrichment of the small amount of NH_4 observed, without assuming bypass flow. Incorporation of bypass flow resulted in increased predicted $\delta^{15}NH_4$-N values (Figure 6), whereas simulated $\delta^{15}NO_3$-N values remained (almost) unaffected. The extreme variation of predicted $\delta^{15}NH_4$ values is explained by the direct infiltration of enriched throughfall water after heavy rainfall into lower soil water compartments with very low NH_4 concentrations. Means of the $\delta^{15}NH_4$-N observations

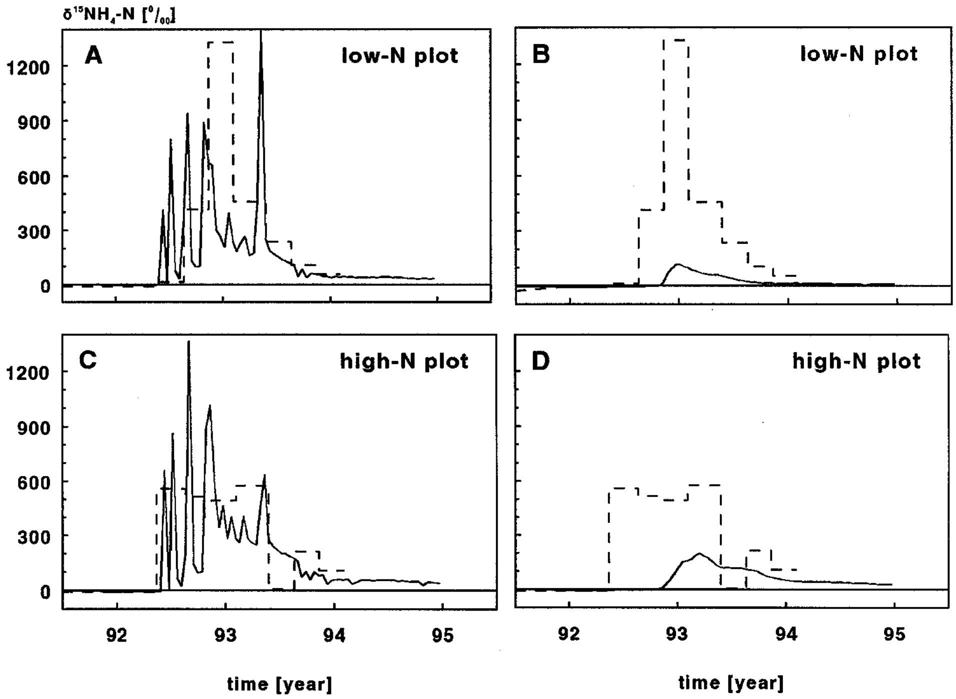

Figure 6. Observed (dashed line) and simulated (solid line) $\delta^{15}NH_4$ values in soil water at 90 cm soil depth in the low-N plot with (**A**) and without (**B**) bypass flow and in the high-N plot with (**C**) and without (**D**) bypass flow.

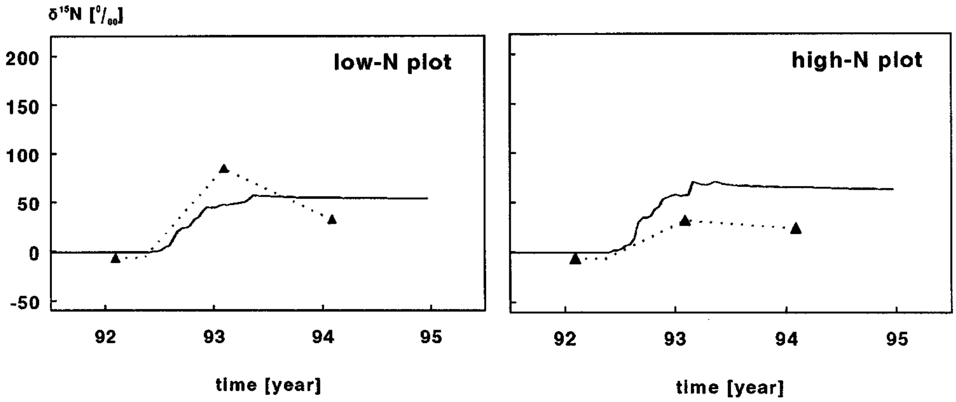

Figure 7. Observed (symbols), model input data (dotted line) and simulated (solid line) $\delta^{15}N$ values of the needles (older than current-year needles) of the low-N and high-N plot at Speuld.

were nearly identical at both treatments (Table IV). Although the time trends were improved, the ME for $\delta^{15}NH_4$-N remained low for both plots.

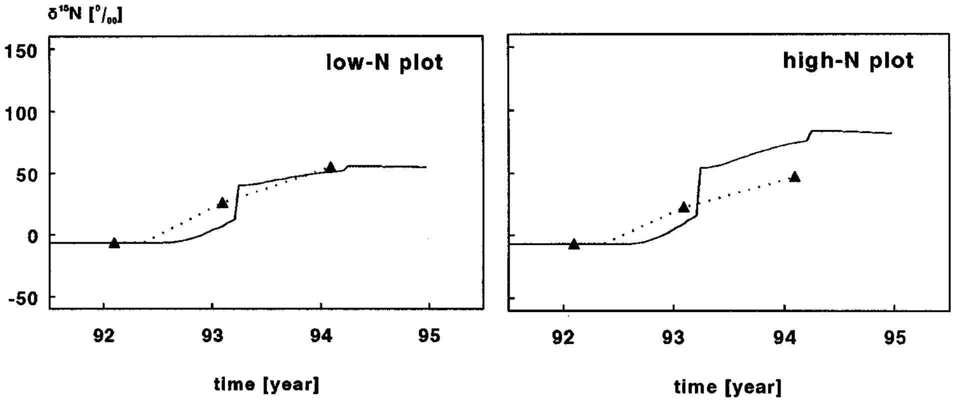

Figure 8. Observed (symbols), model input data (dotted line) and simulated (solid line) $\delta^{15}N$ values of the organic layer at the low-N and high-N plot at Speuld.

The observed and simulated enrichments of the older needles are presented in Figure 7. In the simulations, the current-year needles change into older cohorts at the first of April every year. This leads to the sudden increase in the simulated isotope ratio at that time. However, the older needles sampled in February 1993 indicated that some ^{15}N had reached these older needles by that time already. This indicated a redistribution of N between needle cohorts, a mechanisms originally not included in the NICCCE model. Two percent (2%) of the current-year needle N pool was therefore redistributed into the older needle cohorts every week in the simulations. With the same parameter-set the enrichment of the older needles in the high-N deposition plot was overestimated.

Figure 8 presents the observed and simulated enrichments in the organic layer. In the simulations for the low-N plot, the observed enrichment of the organic layer in 1993 was not reached and overestimated in 1994. In 1994 and 1995 simulated values in the organic layer of the high-N plot where somewhat higher than in the low-N plot. For the high-N plot these simulations were clear overestimates of the observed values.

3.4. Parameters Sensitivity

Sensitive model parameters of the NICCCE model, as quantified by the Kolmogorov d-statistics for the two *cases*, are summarized in Table V. Due to a different number of accepted runs these sensitivities can not be compared directly. However, the two *cases* indicate that changes arise in sensitivity ranking of model parameters if ^{15}N tracer data are added as information to the simulation.

To quantify this change in sensitivity due to information provided with the ^{15}N tracer (*tracer effect*), the Kolmogorov-d was calculated as the maximum difference between cumulative probability distributions of accepted (*case 1*) and 'wrongly accepted' runs (Figure 9). The *tracer effect*, characterized by lower parameter

Table V

Sensitive model parameters ranked according to Kolmogorov d-statistic. A high ranking indicates a relatively high sensitivity of the parameter. *Case 1* refers to simulations without information on the ^{15}N tracer. *Case 2* refers to simulations including the ^{15}N tracer data. *Tracer effect* refers to the impact of the ^{15}N tracer data on parameter sensitivity

Parameter[a]	Unit	Value	Kolmogorov d-statistic		
			Case 1	*Case 2*	*Tracer effect*
frNMetF	(mg mg^{-1})	0.72	0.250	0.311	0.249
Amax	(g C m^{-2} day^{-1})	25.5	0.060	0.127	0.116
Ceff	(mg mg^{-1})	0.57	0.106	0.128	0.107
AmEfHlf	(mg C mg N^{-1})	8.65	0.062	0.091	0.096
NiCNHlf	(mg C mg N^{-1})	10.58	0.106	0.081	0.091
RTRtf	(yr)	1.20	0.055	0.074	0.076
RTCAR	(yr)	0.97	0.051	0.079	0.076
RTSTR	(yr)	1.55	0.032	0.086	0.073
CDisBr	(mg mg^{-1})	0.21	0.084	0.065	0.071
SLA	(cm^2 g^{-1})	58	0.082	0.091	0.071

[a] frNMetF: fraction of microbial N contained in easily decomposable microbial N; Amax: maximum rate of C assimilation per leaf surface area; Ceff: carbon use efficiency; AmEfHlf: C:N ratio of microbial biomass where half of the ammonium is nitrified; NiCNHlf: C:N ratio of microbial biomass where nitrate uptake is half of its maximum value; RTRtf: residence time of fine roots; RTCAR: residence time of carbohydrates; RTSTR residence time of ‘structural’ organic C and N pool; CDisBr: fraction of C assimilated distributed to branches; SLA: specific leaf area.

uncertainties of highly ranked parameters, was in particular found for some parameters describing microbial N transformations (Table V). That lower standard deviations of parameter distribution can be achieved is apparent from the shape of the curve ‘wrongly accepted’ in the example of ‘the fraction of microbial N contained in easily decomposable microbial N’ (Figure 9). The shape of the curve ‘wrongly accepted’ indicates that in particular higher parameter values were wrongly accepted. This change in parameter distribution, observed for several parameters, indicates that uncertainties in parameters can be reduced as a result of ^{15}N tracer data.

4. Discussion

The ^{15}N tracer data contained valuable information for the calibration of the model, particularly of its isotopic part. Although N input-output budgets could be simulated well without using ^{15}N tracer data, parameter distributions changed as a result of the additional information that this ^{15}N tracer data supplied. This was reflected in changes of rankings of parameter sensitivities and a *tracer effect*, which indicates

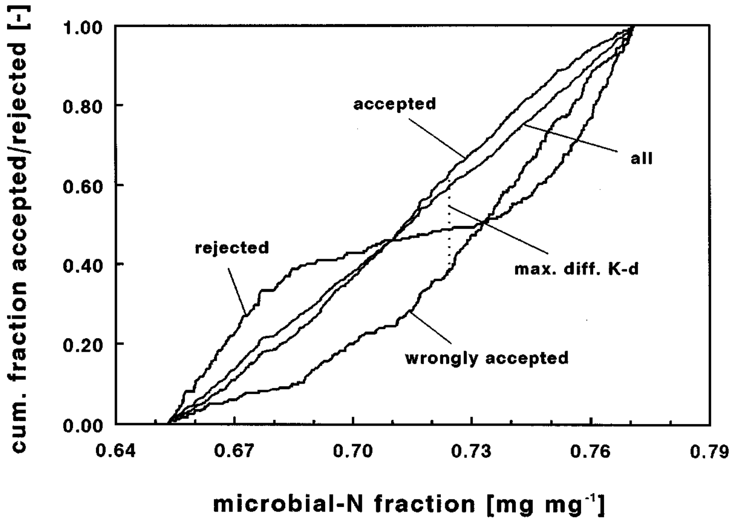

Figure 9. Cumulative probability distributions of the fraction of microbial N contained in easily decomposable microbial N (frNmetF). Runs were accepted if NO_3 concentrations (90 cm depth) and N% of the needles remained within twice the standard deviation of the mean. Runs were wrongly accepted if they were rejected after adding criteria for $\delta^{15}N$ values in NO_3-N (90 cm) and needles. The maximum difference between cumulative probability distributions of accepted and rejected runs is shown (Kolmogorov d-statistic).

how much stronger parameter distributions are confined by incorporation of ^{15}N data (Table V). Although the extra information provided by the tracer data did not necessarily improve the fit of the model to the 'conventional' data, reliability of the model improved: with conventional ^{14}N data only, net N transformations might be simulated with erroneous rates of gross N transformations but equal net rates. It was also found that in using isotope information there was a narrowing of acceptable ranges for parameters describing turnover rates of ecosystem pools particular for those related to mineralization of soil organic matter.

Although the simulation results show that the N transformations in this N saturated ecosystem could be described well by the model NICCCE, interpretations should be handled with care. In our simulations we combined field data from several plots. Uncertainties remain due to variability in N and C pool sizes at the different plots within the ecosystem. In this evaluation we used data from Van der Maas (1990) for initial conditions and for the pre-treatment period. The plots studied by Boxman *et al.* (1995), Koopmans *et al.* (1995) and Koopmans *et al.* (1996) are different with respect to soil conditions, NO_3 and NH_4 concentrations in soil water, and N and C pool sizes of vegetation and soil compartments. These data indicate variability in space and time not accounted for in the model approach. Consequently, the parameter set obtained after calibration of NICCCE cannot be considered as a unique set for describing N and C cycling at Speuld, as it does not account for all these uncertainties.

Table VI

Input (throughfall) and simulated output (leaching at 90 cm soil depth) of water and inorganic N during 4 yr of N manipulations

	Input				Output through leaching			
	H_2O	NH_4	NO_3	Total inorg. N	H_2O	NH_4	NO_3	Total inorg. N
	(mm)	— (kg N ha^{-1} yr^{-1}) —			(mm)	— (kg N ha^{-1} yr^{-1}) —		
Low-N plot								
1991	420	0.3	0.1	0.4	128	0.1	0.7	0.8
1992	646	2.3	1.5	3.8	272	0.3	1.9	2.2
1993	678	5.8	6.4	12.2	523	0.5	4.3	4.8
1994	752	3.5	2.4	5.9	480	0.4	3.1	3.5
Average	624	3.0	2.6	5.6	351	0.3	2.5	2.8
High-N plot								
1991	419	21.2	5.8	27.0	128	0.2	11.4	11.6
1992	644	27.0	9.7	36.7	270	0.3	31.7	32.0
1993	675	34.2	9.7	43.9	471	0.6	37.1	37.7
1994	755	31.2	10.1	41.3	476	0.4	24.1	24.5
Average	623	28.4	8.8	37.2	336	0.4	26.1	26.5

The N input-output budgets for the plots (Table VI) show that the total atmospheric N input at the low-N deposition plot was less than 15% of the atmospheric N input at the high-N plot (Table VI). Total N leaching losses in the low-N plot decreased to about 10% of the leaching losses at the high-N plot, resulting in a leaching loss of about 3 kg N ha^{-1} yr^{-1}. N output as a percentage of N input varied considerably between years from 43 to 87% in the high-N plot. In the first year after reduction of N input, approximately 200% of the very low annual N input leached out of the soil in the low-N plot but ratios between output and input reached levels comparable to the high-N plot in the subsequent years (39 to 59%). The very low N input and output in this low-N plot results in the large variation in output:input ratios found for this plot.

High NO_3-N concentrations observed in soil water, together with the high ^{15}N enrichment of the NO_3 (Figure 5), indicated a rapid nitrification of atmospheric NH_4 in these acid forest soils (Koopmans *et al.*, 1996). Table VII summarizes N fluxes in the organic layer of the low-N and high-N plot, as calculated with the NICCCE model for four successive years of N manipulations. The fluxes should be interpreted with caution, as they depend on process formulations and are sensitive to parameter uncertainties. Total net mineralization was calculated as the transformation of organic N to inorganic N, i.e. gross (NH_4 + NO_3) mineralization

Table VII

Simulated N transformations in the organic layer during 4 yr of N manipulations at Speuld. All transformations in kg N ha^{-1} yr^{-1}

	Gross mineralization		Gross immobilization		Net	Net	Root
	NH_4	NO_3	NH_4	NO_3	mineralization	nitrification	uptake
Low-N plot							
1991	161	25	152	12	22	13	7
1992	182	26	172	13	23	13	8
1993	189	25	180	12	22	13	7
1994	190	22	180	12	20	10	8
Average	181	24	171	12	22	12	8
High-N plot							
1991	172	46	180	18	20	28	9
1992	198	55	209	21	23	34	9
1993	207	58	222	21	22	37	8
1994	204	56	218	22	20	34	9
Average	195	54	207	21	21	33	9

minus gross (NH_4 + NO_3) immobilization. Net nitrification was calculated from gross NO_3 mineralization minus gross NO_3 immobilization. Changes in microbial biomass N were neglected in our calculations. At lower atmospheric N inputs gross N fluxes decreased. In particular, the formations of NO_3-N slowed down from 54 kg N ha^{-1} yr^{-1} to 24 kg N ha^{-1} yr^{-1} which corresponds to the lower gross NH_4-N immobilization. Results revealed similar net mineralization rates in the organic layer under both treatments of about 22 kg N ha^{-1} yr^{-1} but reduced net nitrification rates under lowered N input. These numbers can be compared to net mineralization rates calculated from a year-round field incubation study with closed columns (Koopmans *et al.*, 1995). In this field study a net mineralization rate in the organic layer of respectively 29 and 33 kg N ha^{-1} yr^{-1} was found for the low-N and high-N plot. This difference was not significant. In the same incubation study, a net nitrification rate of 8 kg N ha^{-1} yr^{-1} was found in the organic layer in both treatments. A comparable nitrification rate was found in the simulations for the low-N plot (12 kg N ha^{-1} yr^{-1}), whereas in the high-N plot a much higher net nitrification rate was simulated (33 kg N ha^{-1} yr^{-1}).

The simulations show that at high N inputs all (gross) fluxes are higher (especially for NO_3-N) if compared to the reduced N input levels. A ratio of gross to net N mineralization of 11.9 was found in the high-N plot, whereas this ratio decreased to 9.3 in the low-N plot after the experimental removal of N inputs. These numbers

Table VIII

^{15}N recoveries in the low-N and high-N plots measured in the field (Koopmans *et al.*, 1996), recalculated using N pool sizes from NICCCE and simulated with NICCCE (see text for details)

Ecosystem component	Low-N plot			High-N plot		
	Measured	Recalculated	Simulated	Measured	Recalculated	Simulated
	(%)					
Vegetation	32.7	26.8	19.7	28.8	23.5	24.4
Organic layer	22.2	34.3	48.6	15.0	25.2	40.8
Mineral soil	23.9	16.8	29.5	22.2	15.5	16.6
Leaching	2.1[a]	3.1[b]	2.2[c]	33.1[a]	23.0[b]	18.2[c]
Total recovery	81	81	100	99	87	100

[a] Inorganic N concentrations of composite samples were multiplied by simulated soil water fluxes. [b] Total N leaching losses from the simulations were combined with $\delta^{15}N$ values observed. [c] Fitted to geometric mean inorganic N concentrations in soil water and $\delta^{15}N$ values of composite samples.

correspond to findings of Schimmel *et al.* (1989) and Davidson *et al.* (1990), reporting N turnover through the microbial biomass to be 10–20 times the rate of net N mineralization in soils in native ecosystems but are much higher than the ratio of gross to net N mineralization of less than 3, found in ^{15}N pool-dilution experiments with disturbed soil from the Speuld site in laboratory incubations (Tietema and Van Dam, 1996).

A comparison of the fate of the ^{15}N as calculated directly from measured ^{15}N field data in a mass balance (Koopmans *et al.*, 1996) and calculated from the NICCCE simulations is summarized in Table VIII. In order to evaluate model performance, the ^{15}N budget was recalculated using N pool sizes as they were simulated in NICCCE. Therefore, differences between ‘measured’ and ‘recalculated’ ^{15}N recoveries result from pool size differences between the mass balance approach and the simulations. Differences between the ‘recalculated’ and ‘simulated’ recoveries arise from an inadequate fit of the model to the ^{15}N data. The simulations accounted for roots (part of vegetation pool) in contrast to measured and recalculated ^{15}N budgets. The simulations did not account for heartwood, sapwood and bark separately, as distinguished in the ‘measured’ and ‘recalculated’ ^{15}N budget. Comparing the ‘recalculated’ and ‘simulated’ ^{15}N budget, minor differences in ^{15}N retention of the vegetation were found. Major contradictions were observed for the organic layer, especially for the high-N plot. This followed from the poor model performance for $\delta^{15}N$ values observed for the organic layer (Figure 8). A small difference in $\delta^{15}N$ values results in a considerable discrepancy in the percentage ^{15}N retained as a result of the large N pool in the organic layer. In comparison to the ‘recalculated’ ^{15}N budget of the high-N plot ‘simulated’ ^{15}N recovery in leaching was underestimated by 5% of the ^{15}N applied, in line with the underestimations of $\delta^{15}NH_4$-N and $\delta^{15}NO_3$-N data in soil water at this plot. The results implicate that ^{15}N retention between the organic and mineral layer of the soil is inadequately described by the model and not well understood. Details on NH_4^+ adsorption equilibria, and time series of ^{15}N incorporation into the soil microbial biomass and incorporation in soil organic matter might improve the simulations and increase our knowledge of soil N retention mechanisms.

Our experiments and simulations showed that lowering N inputs to this forest ecosystem resulted in a quick response of inorganic N concentration in the soil water and therefore in the N leaching flux. Results on modelling bypass flow indicated that the interaction between infiltrating throughfall water and soil water was not complete: a small fraction of the incoming water passed through the soil immediately. The simulations indicated that lowering N inputs affected gross N fluxes in particular in this forest soil. A shift occurs towards a diminishing role of NO_3-N in the overall N fluxes.

The ecosystem responses with respect to soil water chemistry, followed by a change in N transformations and nutrient uptake by the vegetation were highly dependent on soil microbial parameters as appeared from the sensitivity analysis. In this study the use of the ^{15}N tracer data contributed to narrow ranges for parameters distributions and decreased parameter uncertainties.

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